

Notes Week 29

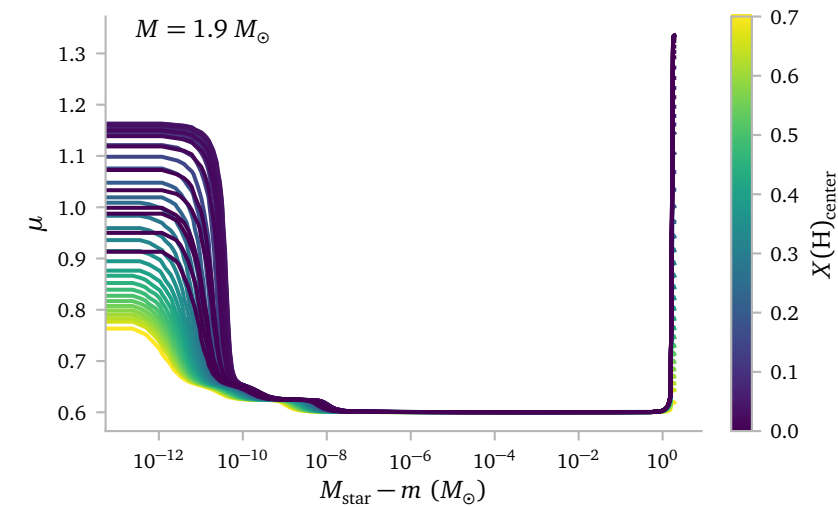
a. Thermohaline Mixing

I simulate a bunch of MS stars with different initial masses. I save profiles depending on the central hydrogen fraction to get an even coverage of the μ profiles throughout the MS evolution of the stars. This is important because the μ profiles change quite a lot throughout the MS evolution. We obviously know the age of the MS star when it accretes mass and thus we can make a pretty good assumption about the mixing.

The problem is then that we could possibly observe the MS stars billions of years after the mass has been accreted. At this point the μ profiles could change and it becomes uncertain.

However, since the abundances are quite uncertain anyway, I think it is okay to just take the μ profile at the time of accretion.

To start, let me plot the μ profile from the envelope to the center.

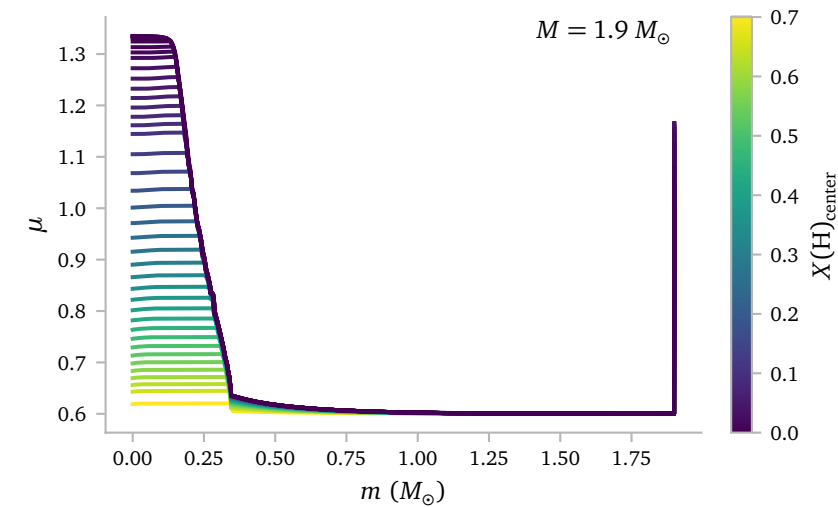


Here, we quite clearly see that outer layers of the envelope, which have very low density, are not fully ionized. This obviously causes the mean molecular weight to increase.

However, I think this is an unimportant effect. Accreted material will just the envelope layer further outwards, and the previously unionized material will become ionized.

The much more interesting part is close to the core.

Let's look at the profile going from the core outwards.

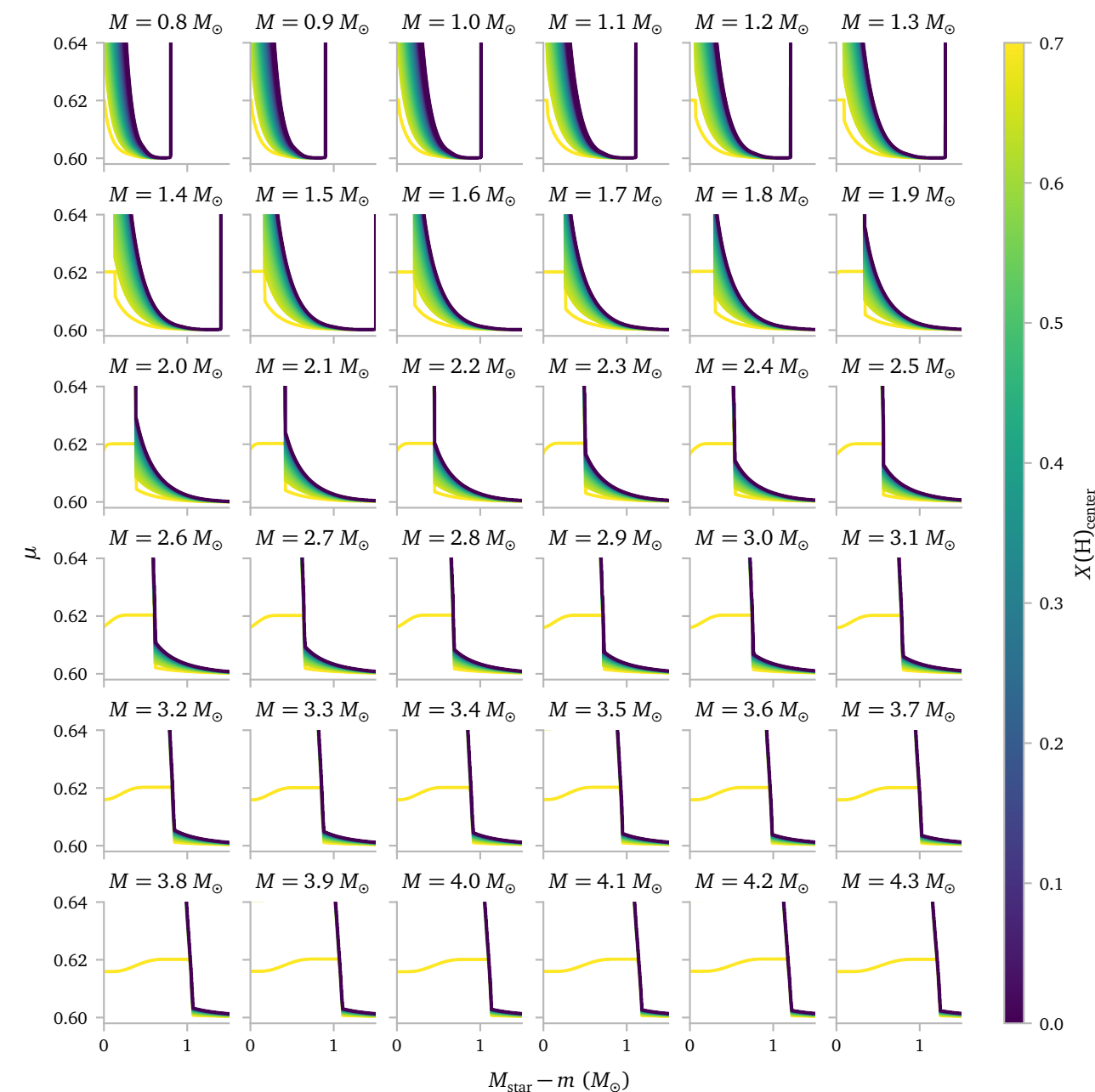


For this particular mass, we see a nearly perfectly diagonal line from the mean molecular weight in the core to the mean mean molecular weight in the envelope.

The diagonal line does not shift with the changing core composition, only the final height changes.

However, just before the “diagonal” line starts, the μ profile already starts to rise a bit. I think this is pretty interesting, as this seems to change quite a bit depending on the core composition, and could thus be important for the moment of accretion.

Below I plot all μ profiles for all models, zooming in on this lower portion.



Clearly, not only the mass position of the rise in μ changes with mass, but also the slow rise before the diagonal line.

However, I have no intuition for the mean molecular weight of some strongly enriched material, so before I do a bunch of calculations, let me make a really simple estimate for μ .

a.1 Simple Derivation

NOTE: Simple derivation for myself to refresh how the mean molecular weight is calculated.

The mean molecular weight is defined as:

$$\mu = \frac{\rho}{nm_u}.$$

Here,

- μ is the mean molecular weight of a gas.
- ρ is the mass density of a gas.
- n is the number density of a gas.
- m_u is the atomic mass unit.

For one element, the mass density contributed is

$$\rho_i = X_i \rho.$$

The number density of nuclei is then simply

$$n_i^{\text{nuc}} = \frac{\rho_i}{A_i m_u}.$$

However, if we assume full ionization, we will have 1 nucleus + Z_i electrons.

The “real” number density is simply

$$n_i = (1 + Z_i) n_i^{\text{nuc}}.$$

Now, let me expand and sum over all elements:

$$\begin{aligned} n &= \sum_i (1 + Z_i) \frac{\rho X_i}{A_i m_u} \\ &= \frac{\rho}{m_u} \sum_i \frac{X_i (1 + Z_i)}{A_i} \\ \frac{nm_u}{\rho} &= \sum_i \frac{X_i (1 + Z_i)}{A_i} \\ \frac{1}{\mu} &= \sum_i \frac{X_i (1 + Z_i)}{A_i} \end{aligned}$$

Cool! I am *assuming* full ionization here, which I think is fine. I am pretty sure nearly the whole MS star is ionized anyway.

Now to compute the mean molecular weight of the accreted gas, we need

- X_i , the mass fraction of every element in the accreted gas. We have this.
- Z_i , the proton number of every accreted element. We also have this.
- A_i , the mass number of every element in the accreted gas. We need to make an *assumption* here.

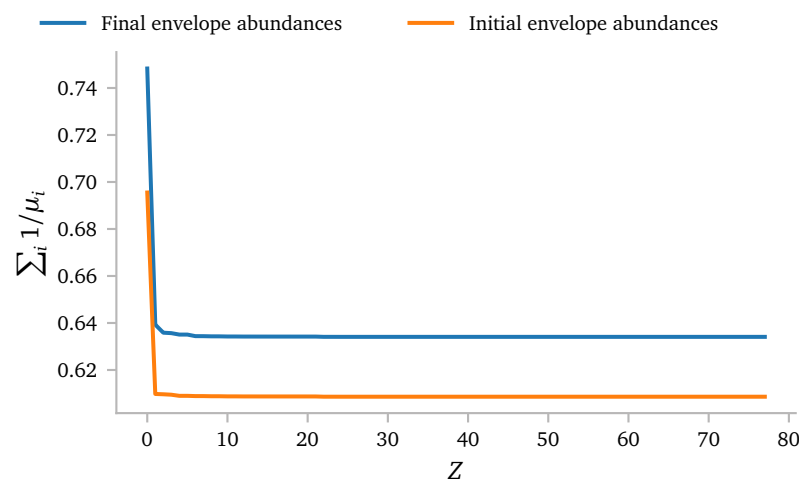
Especially for heavy elements, it is quite possible that the element has a few isotopes with different masses that are all stable and relatively abundant. In this case, the elemental mass should be the averaged mass of the isotopes.

I think it is fine to take the [standard atomic weight](#) in this case. The differences will be small anyway, especially for the heavy elements.

I will use the package `periodictable` written by [Kienzle et al. \(2026\)](#) based on data from [Meija et al. \(2016\)](#).

It is then quite straightforward to μ .

Below I plot the μ I calculated from my MESA model with the Monash Abundances.



I am unsure if the differences in μ between the initial models is due to evolution after the MS, or due to differences between MESA and Monash abundances. But, in the end, the differences for the predicted abundances in the MS star would probably be small.

The idea is now to use this μ and mix the accreted mass, thereby changing μ as well, until the mixed μ is equal to the μ in the star.

There are a couple of important things here.

1. The first elements contribute by far the most to μ , which is to be expected due to their much larger abundances.
2. The μ computed with the final abundances is larger than for the initial envelope abundances.
3. μ computed from the initial envelope abundances is a bit larger than μ in the envelope of the MESA MS models, which have $\mu \approx 0.60$.

a.2 Simple Derivation 2

To see how this mixing would work, let me look at the definition of μ again.

$$\mu = \frac{\rho}{nm_u} = \frac{m/V}{Nm_u/V} = \frac{m}{Nm_u}.$$

Say we mix m_1 of gas with mean molecular weight μ_1 with m_2 of a gas with μ_2 . We want to know what the resulting μ is of the mixed whole.

We would calculate N using μ and m to obtain:

$$N_i = \frac{m_i}{\mu_i m_u} \implies \mu_{\text{mix}} = \frac{m_1 + m_2}{(N_1 + N_2)m_u} = \frac{m_1 + m_2}{\frac{m_1}{\mu_1} + \frac{m_2}{\mu_2}}$$

Or, more conveniently:

$$\mu_{\text{mix}}^{-1} = \frac{m_1 \mu_1^{-1} + m_2 \mu_2^{-1}}{m_1 + m_2}, \quad m_{\text{mix}} = m_1 + m_2.$$

I think it should be pretty easy to implement this logic for simulating the dilution of μ as the accreted material starts to mix.

Here is a minimal working example:

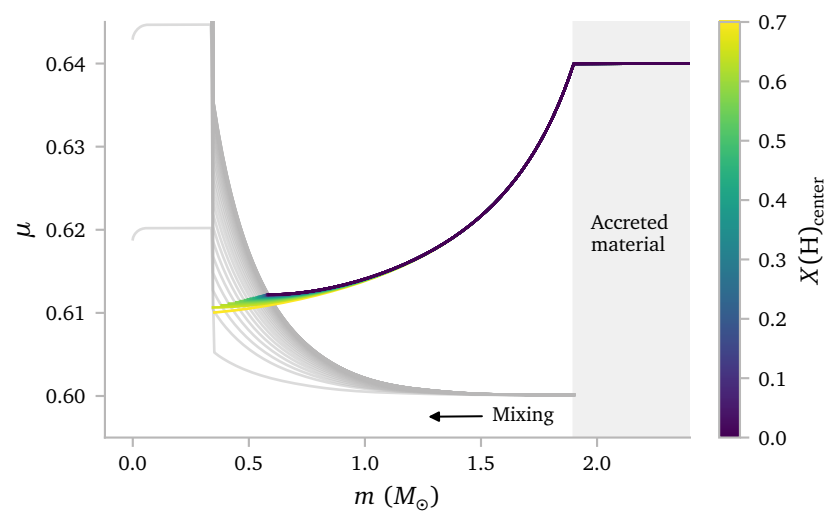
```
def effective_mu(profile, M_acc, mu_acc):
    ms = -np.diff(profile.mass)
    mus = (profile.mu[1:] + profile.mu[:-1]) / 2
    arg = np.argmin(mus)
    mus[:arg] = mus[arg]
    mu_current = mu_acc
    m_current = M_acc

    mu_profile = [mu_acc]

    crossing_index = None
    for m, mu in zip(ms, mus):
        mu_last = mu_current
        mu_current = (m + m_current) / ((m / mu) + (m_current / mu_current))
        m_current = m + m_current
        mu_profile.append(mu_current)

    mus = np.concatenate([[mus[0]], mus])
    mu_profile_original = mus
    return mu_profile, mu_profile_original
```

Below I show an example of $0.5 M_\odot$ being accreted by a $1.9 M_\odot$ MS star at various times throughout the MS.

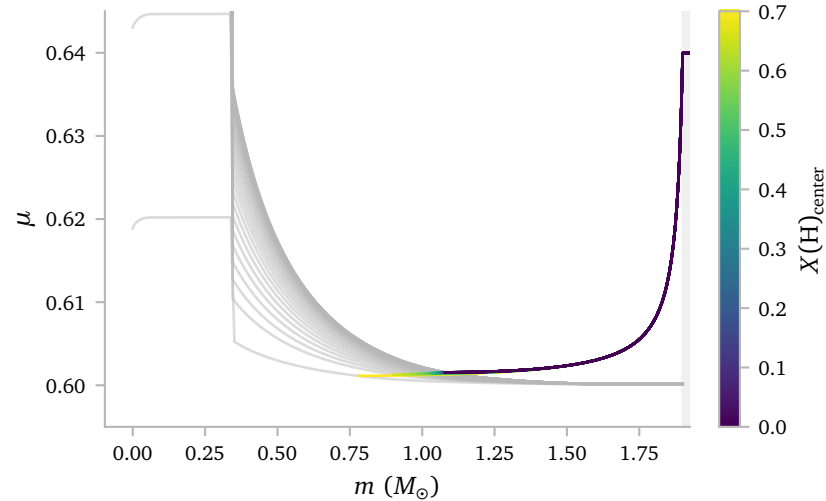


Here, accreted mass gets added to the top of the envelope of the original star.

As mixing extends into the envelope, the mean molecular weight of the mixed material decreases from the initial accreted mean molecular weight to lower values until at some point, the mean molecular weight of the mixed material is *lower* than the mean molecular weight of the envelope of the original star.

This critical point roughly determines how deep mixing extends.

To me, this seems like it works as expected. Another test would be to see how mixing works with much less accretion. In this case, mixing should cause the mean molecular weight to drop much quicker.



This plot show the same $1.9 M_\odot$ model accreting “only” $0.025 M_\odot$ of strongly enriched mass. And indeed, we see a much quicker drop-off and *less* mixing.

We also see that the age of the MS star does change the mixing depth a bit. In both case I have shown here, the mixing depth decreases with age.

This is a nice property, because by using the μ profile at the time of accretion, we are making an *optimistic* estimate for the amount of mixing, and thus a *conservative* estimate for the abundances.

Now we are quite easily able to obtain the mixing mass. In combination with the amount of accreted mass and the mass fraction of the elements in the accreted mass, we are almost ready to compute the envelope abundances in the main sequence stars.

The last thing we need is the envelope abundances of every element in the envelope of the MS star. I think this is a bit difficult.

a.3 Interpolated mean molecular weight calculations.

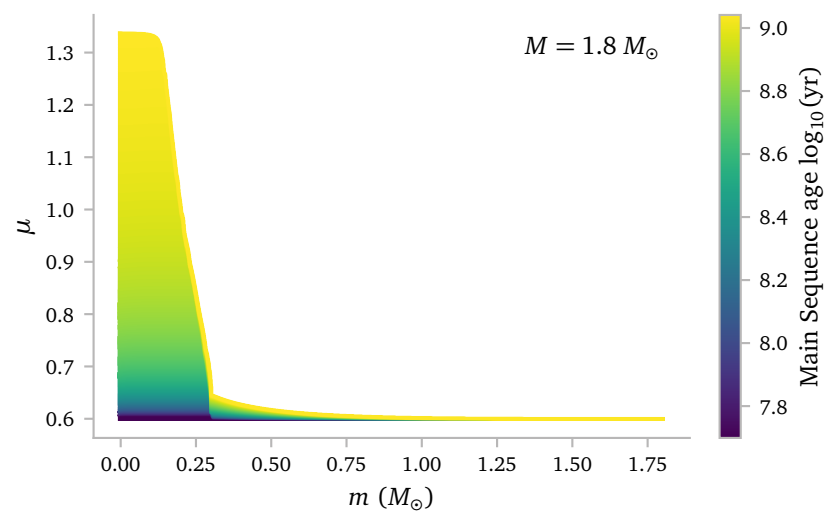
The basic idea is very similar to the interpolation of the abundances. First, I interpolate the μ profiles for different ages¹. Then I interpolate these profiles between the masses.

a.3.1 Main sequence age to central Hydrogen composition.

This is just a simple linear interpolation as well.

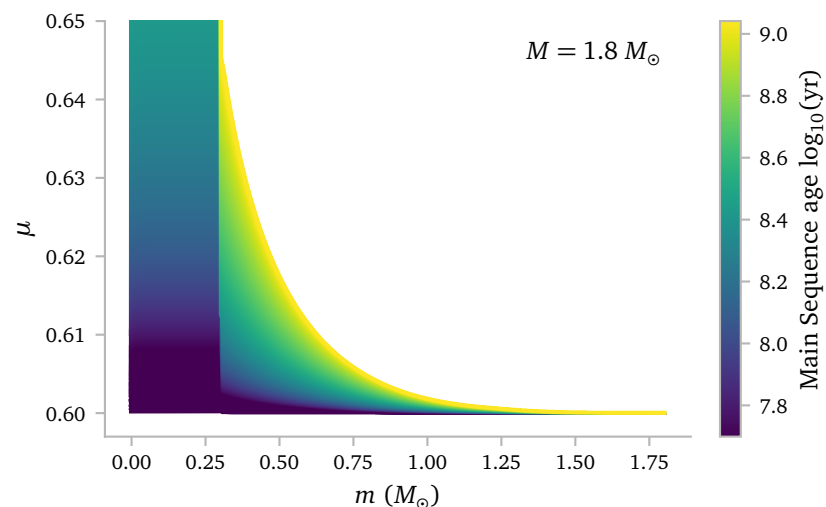
a.3.2 Interpolated Age and Mass

I have now created some code that can quickly generate interpolated profiles. Below, I show two examples. In the first example, I change the age of a MS star to see how the μ profile changes.



We can clearly see that, as the star becomes older, the mean molecular weight of the star in the core rises. This is to be expected as hydrogen turns into helium.

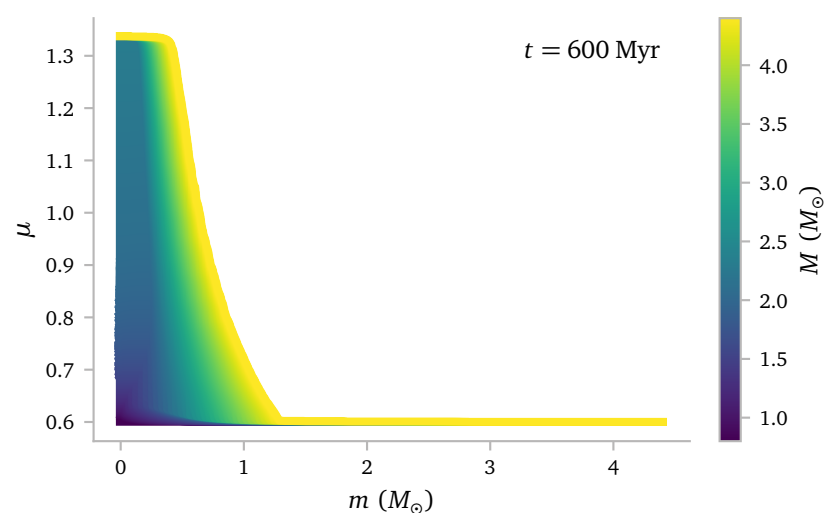
We can also see that the slow sharp rise happens at the same masses for all ages. However, the slow rise before the sharp rise to the core mean molecular weight value does depend on age. In the next figure, I zoom in on this.



The zoom very clearly shows the sharp boundary between the slow and fast rise of the mean molecular weight *and* the effect of age.

This is really the regime that we are interested in, because the mean molecular weight of the accreted mass will in all cases be pretty close to my estimated value of 0.64.

We can also look different masses at a constant age.



Now, we can very clearly see that the mass position of the rise in molecular weight changes with mass. This makes perfect sense. It is really just a proxy for the core mass of the star anyway, and lighter stars will have lighter cores.

Since we are comparing stars of different masses at the same age, we also see that the maximum μ in the core increases with mass too. This also makes perfect sense, since heavier stars have shorter main-sequence lifetimes, and thus, at the same age will have burned through more of their core hydrogen.

References

Kienzle, P., Maranville, B. B., Cooper, J., et al. (2026) *python-periodictable/periodictable: R2.1.0 2026-02-27*.

Meija, J., Coplen, T. B., Berglund, M., et al. (2016) *Atomic weights of the elements 2013 (IUPAC Technical Report)*. *Pure and Applied Chemistry*88, 265.